

Agentic AI for Clinical Decision-Making

Autonomous Authorization and Risk Assessment in Behavioral Healthcare

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Clinical authorization is a sequential evidence problem

Continued-stay authorization in behavioral healthcare requires clinicians to evaluate patient information scattered across **11 disparate clinical data sources** — including the C-SSRS, PHQ-9, and nursing narratives — to render a defensible approve-or-deny decision.

The standard LLM pattern loads every document into context before deciding. Cost then scales with the **size of the record**, not the **difficulty of the decision**.

In practice, many cases are decidable long before all evidence is read. What has been missing is a defensible rule for *when an autonomous agent has read enough*.

RESEARCH QUESTION

Can an autonomous agent choose *which source to read next* — and stop the moment the evidence is statistically sufficient?

Three stopping criteria

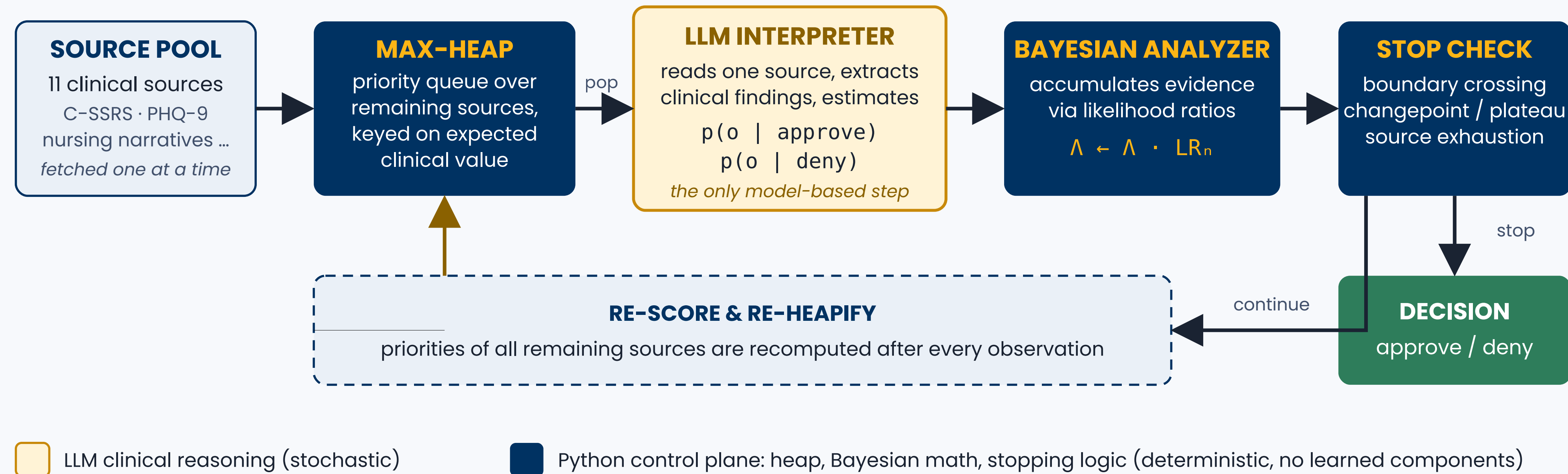
- Bayesian boundary crossing**
The cumulative likelihood ratio exits the continue region [B, A].
- Changepoint detection**
Additional sources stop moving the evidence — a plateau is detected.
- Source exhaustion**
All 11 sources read: the fallback is equivalent to the full-context baseline.

Clinical source categories

Risk assessment C-SSRS, suicide-risk patterns, direct quotes	Symptom severity PHQ-9, GAD-7, PCL-5
Narrative context intake, mood, sleep, denial patterns, nursing notes	Treatment context medication adherence

The heap selects the next most informative source across categories, given current evidence.

Architecture: LLM interprets, Python decides

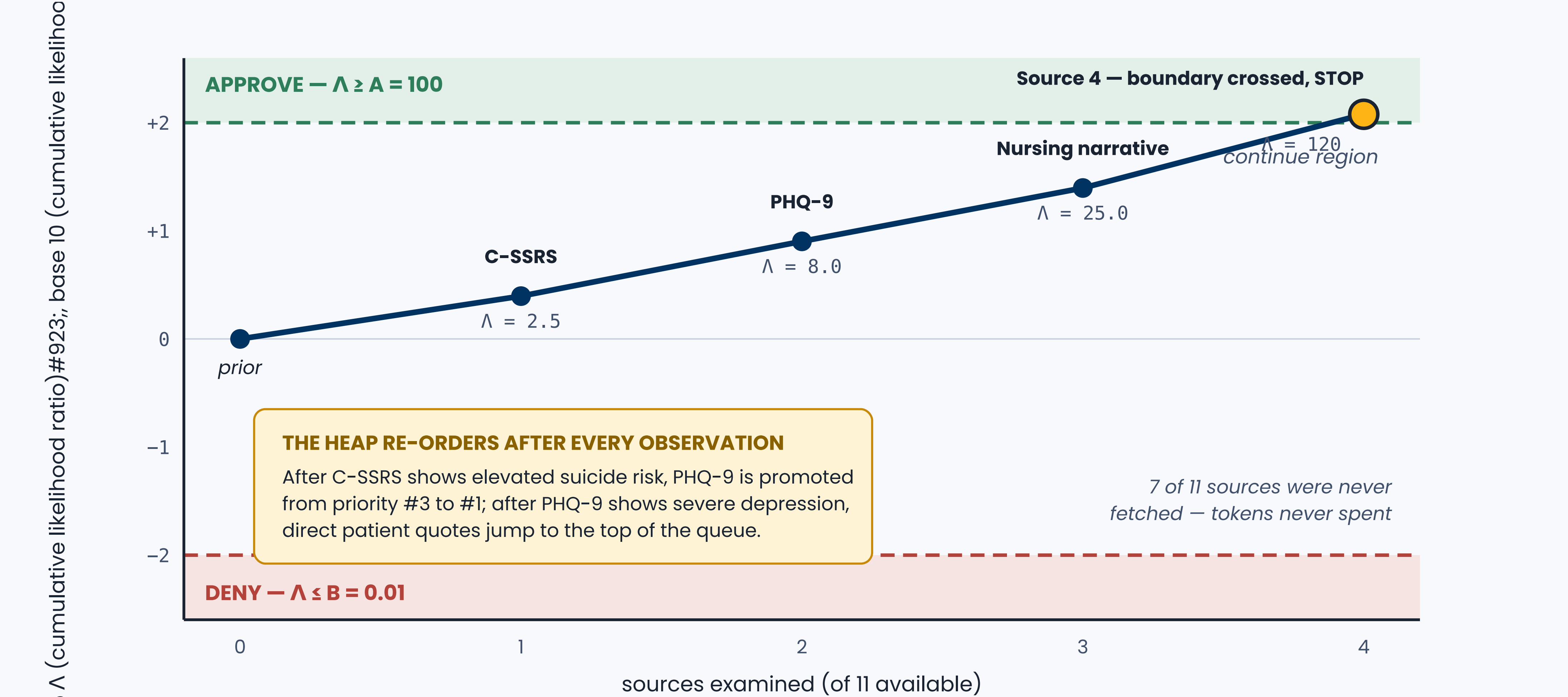


$$\Lambda_n = \prod_{i=1..n} p(o_i | H_1) / p(o_i | H_0) \quad H_1: \text{approve extension} \quad H_0: \text{deny}$$

Stop & **approve** when $\Lambda_n \geq A \approx (1-\beta)/\alpha$ · stop & **deny** when $\Lambda_n \leq B \approx \beta/(1-\alpha)$ · otherwise continue

SPRT minimizes expected observations at fixed error rates for i.i.d. observations (Wald-Wolfowitz); applied here as a principled stopping rule over heterogeneous clinical sources.

Sequential execution: stop when the evidence crosses the boundary



Illustrative execution trace based on the production system's worked example; boundaries shown for $\alpha = \beta \approx 0.01$. The system also halts on changepoint-detected evidence plateau or source exhaustion.

Division of labor

- LLM — stochastic clinical reasoning**
scores candidate sources · interprets structured & narrative documents · extracts findings · estimates $p(o | \text{approve})$, $p(o | \text{deny})$
- Python — deterministic control**
heap priorities · cumulative evidence state · likelihood-ratio updates · boundary & plateau checks · stopping loop

Deployment insights

- Deployed in production at a large behavioral healthcare provider.
- High-confidence decisions frequently reached after only **2-5 of 11** sources.
- Significant token and cost efficiency; reviewer agreement tracked in production.

Takeaways

- Gather evidence sequentially; stop when **statistically sufficient** — not when sources run out.
- A clean separation: LLM clinical reasoning vs. Python-controlled orchestration, Bayesian math, and stopping logic.
- Real-world deployment lessons from autonomous clinical authorization.
- Transferable beyond behavioral health: swap the hypothesis space, source pool, and likelihood interpreter.

Constraints & limitations

- Provider identity and exact deployment metrics withheld for confidentiality.
- Presents a reusable architecture and lessons, not proprietary code.
- Bayesian thresholds require domain calibration, human oversight, and policy alignment.